

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 6-2025: A 62-Year-Old Man with Abdominal Pain

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PRESENTATION OF CASE

Dr. Brian Y. Chang (Medicine): A 62-year-old man was admitted to this hospital because of abdominal pain.

Two years before the current presentation, postprandial abdominal pain developed. One year before the current presentation, while living in Costa Rica, the patient received a diagnosis of cirrhosis. Eight months later, he immigrated to the United States. The medical teams caring for him in the United States did not have access to records from his medical care in Costa Rica, and the details of how the diagnosis of cirrhosis had been made were not available to them.

Three months before the current presentation, the patient noted a reducible umbilical hernia. Six weeks later, postprandial abdominal pain worsened in severity and frequency and was associated with nausea. One month before the current presentation, the patient was unable to reduce the umbilical hernia; he presented to another hospital for evaluation. The blood levels of electrolytes, glucose, lipase, and lactic acid were normal, as were the results of tests of kidney function and coagulation; other test results are shown in Table 1.

Computed tomography (CT) of the abdomen and pelvis, performed after the administration of intravenous contrast material, showed a paraumbilical hernia with a 3.6-cm neck that contained a nonobstructed loop of small bowel (Fig. 1). The liver had diffuse hypoattenuation with a lobulated contour and shrunken appearance. Extensive paraesophageal and mesenteric varices were present; the portal vein and superior mesenteric vein were not identified, a finding that suggested cavernous transformation of the portal vein. The spleen measured 18.2 cm in the craniocaudal dimension (reference value, <13.0 cm). Doppler ultrasonography of the abdomen showed no flow in the main portal vein and normal flow through the patent hepatic veins, hepatic artery, and inferior vena cava.

On examination by the general surgery consultant, the hernia was easily reducible, and no surgical intervention was planned for this patient. However, he was admitted to the hospital for further evaluation of abdominal pain. The blood levels

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Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Other Hospital	4 Wk before Current Presentation, Other Hospital	Reference Range, This Hospital†	On Current Presentation, This Hospital
Hematocrit (%)	41.0–53.0	42.2	41.0–53.0	40.5
Hemoglobin (g/dl)	13.5–17.5	14.5	13.5–17.5	14.0
White-cell count (per μ l)	4000–11,000	3410	4500–11,000	3900
Differential count (per μ l)				
Neutrophils	1800–7000	2270	1800–7700	1740
Lymphocytes	1000–4800	730	1000–4800	1400
Monocytes	200–1200	320	200–1200	580
Eosinophils	50–250	60	0–900	110
Basophils	0–300	20	0–300	20
Platelet count (per μ l)	150,000–450,000	60,000	130,000–400,000	70,000
Alkaline phosphatase (U/liter)	40–130	141	45–115	118
Alanine aminotransferase (U/liter)	10–50	48	10–55	43
Aspartate aminotransferase (U/liter)	15–41	72	10–40	53
Total bilirubin (mg/dl)	0.0–1.2	1.5	0.0–1.0	1.8
Direct bilirubin (mg/dl)	0.0–0.2	0.5	0.0–0.4	0.5
Partial-thromboplastin time (sec)	—	—	22.0–36.6	29.6
International normalized ratio	0.8–1.2	1.1	0.9–1.1	1.1
Total protein (g/dl)	6.1–8.1	7.7	6.0–8.3	8.7
Albumin (g/dl)	3.5–5.2	4.1	3.3–5.0	4.5
Lactic acid (mmol/liter)	0.5–2.0	1.5	0.5–2.0	2.1

* To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

of ferritin and iron were normal; the transferrin saturation was 62% (reference range, 14 to 50) and the iron-binding capacity 165 μ g per deciliter (29.5 μ mol per liter; reference range, 250 to 400 [44.8 to 71.6]). Screening tests for hepatitis B and C virus antibodies were negative, and nucleic acid testing did not detect hepatitis C RNA. Additional tests for possible causes of cirrhosis were negative, including serum protein electrophoresis, measurement of ceruloplasmin and α_1 -antitrypsin, and screening for human immunodeficiency virus infection. Antinuclear antibodies, anti-smooth muscle antibodies, antimitochondrial antibodies, and antineutrophil cytoplasmic antibodies were not detected. The blood level of α -fetoprotein was normal. Treatment with pantoprazole was started. There was no hematemesis or melena, and the blood

level of hemoglobin remained stable. Abdominal pain persisted, but the patient was able to eat a regular diet, and he was discharged on hospital day 3 with plans for outpatient esophagogastroduodenoscopy (EGD) and colonoscopy.

Ten days before the current presentation, EGD revealed grade 3 (large) nonbleeding esophageal varices, as well as portal hypertensive gastropathy; colonoscopy revealed nonbleeding internal hemorrhoids and rectal varices. Treatment with nadolol was started, and evaluation in the gastroenterology clinic of the other hospital was planned for further evaluation and management of a presumed diagnosis of cirrhosis.

On the day of the current presentation, postprandial abdominal pain increased in severity and did not abate; the patient presented to the emergency department of this hospital for evaluation.

He reported pain after eating, with less pain after eating bland, low-fat foods. Nausea and vomiting occurred during these periods of postprandial abdominal pain. He estimated a weight loss of 20 kg over the previous 2 months. No swelling was observed in the legs or abdomen, and there were no episodes of confusion. He had had hepatitis as a child, which had resolved without treatment. Other medical history included a motor vehicle crash 5 years before the current presentation in which the patient had been an unrestrained driver and sustained abdominal trauma from the steering wheel of the car. He had no surgical history. Medications included pantoprazole and nadolol; he had no known allergies to medications. The patient had worked in cattle farming while living in Costa Rica and had not traveled outside of Costa Rica before immigrating to the United States. He did not drink alcohol, smoke tobacco, or use illicit drugs. There was no family history of liver disease or thrombosis.

On examination, the temporal temperature was 36.7°C, the pulse 49 beats per minute, the

blood pressure 90/61 mm Hg, and the oxygen saturation 97% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 26.7. He was alert and oriented; there was no asterixis. The sclerae were anicteric. The abdomen was diffusely tender on palpation, with pain that was worse in the upper abdomen, without rebound or guarding. No spider angiomas were observed. The remainder of the examination was normal.

Results of laboratory tests are shown in Table 1. Additional diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Gabrielle K. Bromberg: I participated in the care of this patient and am aware of the final diagnosis. This 62-year-old man, who had immigrated to the United States from Costa Rica, had a 2-year history of postprandial abdominal pain and presented with persistent pain associated with nausea, vomiting, and weight loss.

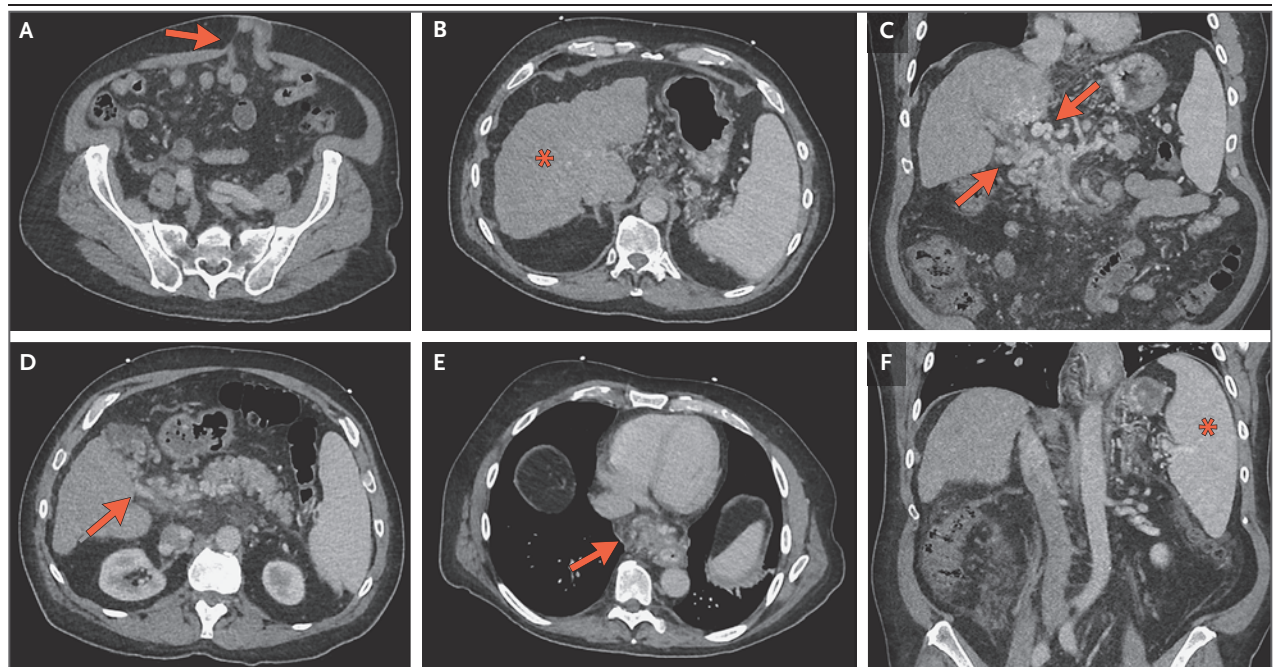


Figure 1. CT of the Abdomen and Pelvis before Admission.

A contrast-enhanced axial image at the level of the umbilicus (Panel A) reveals an umbilical hernia containing a nonobstructed loop of small bowel (arrow). An axial image at the level of the upper abdomen (Panel B) reveals a shrunken liver with lobulated contour (asterisk). Coronal and axial images through the porta hepatis (Panels C and D, respectively) show a poorly defined main portal vein and extensive collateral vessels (arrows), findings that are consistent with cavernous transformation. In an axial image (Panel E), esophageal varices are visible in the lower thorax (arrow). A coronal image through the long axis of the spleen (Panel F) reveals an enlarged spleen (asterisk).

There was a reported diagnosis of cirrhosis complicated by portosystemic shunts, splenomegaly, and portal vein thrombosis (PVT) with cavernous transformation.

On admission to this hospital, years of diagnostic momentum suggested that cirrhosis of the liver was the unifying diagnosis responsible for the patient's weight loss, abdominal pain, portal hypertension, and secondary PVT. However, there are several discordant findings to reconcile, including the presence of splenomegaly and extensive varices and the normal international normalized ratio (INR) and albumin level. If this patient had cirrhosis, hepatic fibrosis would be expected to cause both flow resistance leading to portal hypertension and impaired synthetic function.¹

The inexplicably normal INR and albumin level in a patient with no identifiable risk factors for cirrhosis and normal test results for causes of cirrhosis increases the possibility that cirrhosis was not the underlying cause of this patient's illness. At this point, I would pause to question our assumptions and unveil the missing pieces.²⁻⁴

PORTAL HYPERTENSION

We will begin with something we know unequivocally: this patient's imaging revealed a substantial burden of portal hypertension with portosystemic shunts. Portal hypertension, characterized by high pressure in the portal vein that results in increased resistance to portal blood flow, has many causes.^{5,6} Flow resistance can develop in any segment of the perihepatic vasculature and is divided into posthepatic, intrahepatic, and prehepatic causes.⁵ Intrahepatic causes of portal hypertension can be further divided into presinusoidal, sinusoidal, and postsinusoidal causes on the basis of the location of flow obstruction within the liver.⁵

In Western countries, 90% of cases of portal hypertension are caused by cirrhosis, with only 10% of cases having a noncirrhotic cause.⁵ Globally, noncirrhotic portal hypertension is more prevalent than in Western countries, with infections such as hepatic schistosomiasis affecting more than 200 million people worldwide.^{5,7,8} Although cirrhosis remains a common cause of portal hypertension globally, cirrhosis too has substantial regional variation in its causes; viral hepatitis has varied prevalence, and disparities persist in vaccination coverage and access to treatment.⁸

The differential diagnosis for portal hypertension is broad and includes many rare causes for which this patient had no known risk factors.^{5,9} In constructing a differential diagnosis, I will focus on more common processes that are not readily ruled out by the exposure history, laboratory testing, and imaging.

Posthepatic Causes

Posthepatic portal hypertension can be caused by an inferior vena cava clot or extrinsic compression (ruled out with imaging) and cardiac processes, such as constrictive pericarditis and restrictive cardiomyopathy. This patient did not report any classic symptoms of these processes, such as dyspnea, edema, or decreased exercise tolerance. Furthermore, he had no history of previous pericarditis that would suggest an increased risk of constrictive pericarditis, and the lower thoracic CT images did not reveal pulmonary edema, pericardial effusion, or other intrathoracic abnormalities. Constrictive pericarditis and restrictive cardiomyopathy could not be definitively ruled out but are unlikely potential causes of portal hypertension in this patient.

Intrahepatic Causes

Cirrhosis is the most common cause of intrahepatic portal hypertension. Although cirrhosis was previously assumed to be the underlying process in this patient, other intrahepatic disorders should be considered.

Hepatic amyloidosis commonly manifests with an elevated alkaline phosphatase level and hepatomegaly. Hepatic sarcoidosis also causes an elevated level of alkaline phosphatase and is most often asymptomatic. When hepatic sarcoidosis manifests clinically, levels of aspartate aminotransferase and alanine aminotransferase are elevated, and imaging reveals hepatic nodules and hepatomegaly. This patient had none of these laboratory or imaging findings, which makes an infiltrative hepatic process an unlikely diagnosis.

Hepatic schistosomiasis is most often caused by the *Schistosoma mansoni* or *S. japonicum* species of this parasitic blood fluke. In patients with hepatic schistosomiasis, schistosomes deposit eggs in the portal vein, which leads to disease manifestations. Eggs lodge in the presinusoidal periportal space. In patients with acute hepatic schistosomiasis, nonfibrotic granulomas form.

In chronic cases, inflammation drives periportal fibrosis, and ultimately, portal hypertension develops. In the classic form of schistosomiasis, hepatocellular function is preserved, which is consistent with the normal albumin level and INR in this patient. However, his exposure history did not fit with this diagnosis.^{7,8} Of all schistosomiasis cases worldwide, 93% occur in Ghana, Tanzania, Nigeria, Mozambique, and the Democratic Republic of Congo.⁸ Schistosomiasis is not endemic to Costa Rica, and this patient had no other travel history.¹⁰

Prehepatic Causes

Prehepatic causes of portal hypertension can include extrinsic compression of the portal venous system, as well as PVT. Extrinsic compression was ruled out with imaging.

PVT is a consequence of portal hypertension, but it can also be a cause. Instead of assuming that this patient's PVT resulted from hepatic fibrosis driving venous stasis, a broader differential diagnosis for PVT should be considered.¹¹⁻¹³ Many frameworks for this differential diagnosis branch from the presence or absence of cirrhosis, which is unclear in this patient. Therefore, the other two factors in Virchow's triad (hypercoagulability and endothelial injury) should be examined to identify other potential causes of PVT.

Hypercoagulability can be inherited, as in gene mutations and protein deficiencies, or acquired. Acquired hypercoagulable states can be local or systemic and are caused by inflammation, infections, or cancer. This patient had no known inflammatory bowel disease, pancreatitis, antiphospholipid antibody syndrome, abdominal infections, or cancer, and he did not receive systemic hormone therapy.

Although this patient had no history of abdominal surgery, he had been involved in a motor vehicle crash in which he was unrestrained and sustained abdominal trauma from the steering wheel 5 years before the current presentation. Trauma-related endothelial injury could be a nidus for PVT formation. This degree of chronicity of the PVT was supported by the finding of portal venous cavernous transformation on imaging.¹¹

HEPATIC ATROPHY

If the PVT was caused by endothelial trauma from the motor vehicle crash, and not by cirrho-

sis, then why did the liver appear shrunken on imaging? The liver's unique blood supply — 75% of which is derived from deoxygenated, large-volume portal venous flow — carries essential, hepatotrophic substances from the postintestinal splanchnic vasculature.¹⁴ It has been known since the days of Pavlov that decreased portal venous delivery of essential hepatotrophic substances leads to hepatic atrophy.¹⁵⁻¹⁷

The development of a fully occlusive PVT after the motor vehicle crash 5 years earlier would have led to a substantial reduction in hepatotrophic substance-rich portal blood supply. Over time, this decrease would result in hepatic atrophy and a shrunken liver on imaging. The fully occlusive PVT would lead to proximal portal hypertension. The progressive hepatic atrophy could independently lead to worsened portal hypertension over time through intrahepatic, extravascular, and structural microcirculatory distortion, although nonatrophic hepatocytes would retain their synthetic function — a feature that differs from the pathophysiology of cirrhosis.⁵ Both the PVT itself and the hepatic atrophy would contribute to the portal hypertension and portosystemic shunting that was seen in this patient.

ABDOMINAL PAIN

Without anticoagulation, proximal PVT propagation into other mesenteric veins could produce intestinal venous congestion. Postprandial augmentation of splanchnic flow would cause pain to intensify and drive weight loss. To establish the diagnosis of hepatic atrophy due to chronic PVT, I would perform a liver biopsy. To evaluate for clot extension into the mesenteric veins, I would repeat abdominal imaging.

DR. GABRIELLE K. BROMBERG'S DIAGNOSIS

Hepatic atrophy due to chronic portal vein thrombosis, with probable mesenteric congestion causing postprandial abdominal pain.

DIAGNOSTIC TESTING

Dr. Robert G. Rasmussen: CT of the abdomen and pelvis, performed after the administration of intravenous contrast material during the portal venous phase, revealed a markedly narrowed

main portal vein with evidence of thrombus formation (Fig. 2). The thrombus was observed to extend from the main portal vein to the portosplenic confluence and farther into the superior mesenteric vein. There was no evidence of mesenteric ischemia.

Dr. Jonathan N. Glickman: A transjugular liver biopsy showed intact cores of liver parenchyma with mild distortion of lobular architecture and expanded portal tracts (Fig. 3). Trichrome staining highlighted portal tract fibrous expansion but did not show bridging fibrosis or cirrhosis. The portal areas contained multiple irregular vascular profiles, probably representing portal veins, some of which showed medial hyperplasia or intravascular thrombi. Within the portal areas, the hepatic arteries were normal, and bile duct profiles were present without injury or ductular reaction. Reticulin staining highlighted lobular parenchyma in a vaguely nodular pattern of expansion with compression and atrophy of hepatic plates (Fig. 3D). No sinusoidal congestion or central venous changes indicative of outflow tract obstruction were present. The portal and lobular changes were consistent with parenchymal atrophy, without evidence of advanced fibrosis. There was no pathologically significant inflammation or steatosis to suggest an additional underlying cause of liver disease. Iron staining was negative for stainable iron, and periodic acid–Schiff staining was negative for intracytoplasmic globules.

PATHOLOGICAL DIAGNOSIS

Portal vascular remodeling, with features consistent with the effects of portal vein thrombosis. Nodular regenerative hyperplasia–like changes in hepatic parenchyma.

Hepatic parenchymal distortion, atrophy, and fibrosis, with no features of cirrhosis.

DISCUSSION OF MANAGEMENT

Dr. Kenneth E. Sherman: The management of PVT depends on whether the process is acute (<6 months) or chronic (≥6 months) and whether cirrhosis is present.¹³ In this patient, both history and histologic features classified the process as chronic and noncirrhotic. In such cases, initial management involves upper endoscopy to determine the presence of esophageal and gastric varices, followed by variceal banding with or without the use of nonselective beta-blockers to reduce splanchnic pressures. Both variceal banding and beta-blockers were used in the treatment of this patient. Decisions about anticoagulation are based on the risk of clot extension and the presence of underlying factors that might support a thrombotic state. Hence, we sought input from the hematology service of this hospital. Finally, efforts to reduce portal pressure by shunting were considered. A variety of surgical shunt techniques are available but have rarely been used since the development of the transjugular intrahepatic portosystemic shunt (TIPS),

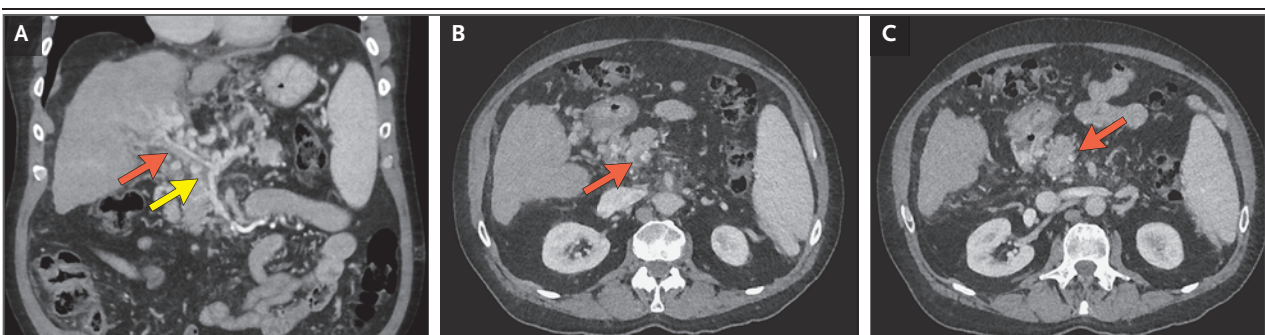


Figure 2. CT of the Abdomen and Pelvis after Admission.

Contrast-enhanced CT was performed in the portal venous phase. An oblique coronal image through the portosplenic confluence (Panel A) shows a diminutive main portal vein with a thrombus (orange arrow). The thrombus extends inferiorly beyond the portosplenic confluence into the superior mesenteric vein (yellow arrow). An axial image through the superior mesenteric vein just inferior to the portosplenic confluence (Panel B) shows the thrombus along the lateral aspect of the superior mesenteric vein lumen (arrow). An axial image more inferiorly at the level of the left renal vein (Panel C) shows the diminutive superior mesenteric vein with the thrombus along the antero-medial aspect (arrow).

which is placed by interventional radiologists. Most shunt procedures divert flow (portal flow is shunted to the systemic circulation), but some surgical and interventional radiology procedures may allow preservation of portal flow to the hepatocytes.¹⁸ We consulted the interventional

radiology service of this hospital to see if one of these options would be possible.

Dr. Sanjeeva P. Kalva: Given the patient's symptoms, portal vein recanalization and TIPS placement (PVR-TIPS) were considered along with angioplasty of the superior mesenteric vein and

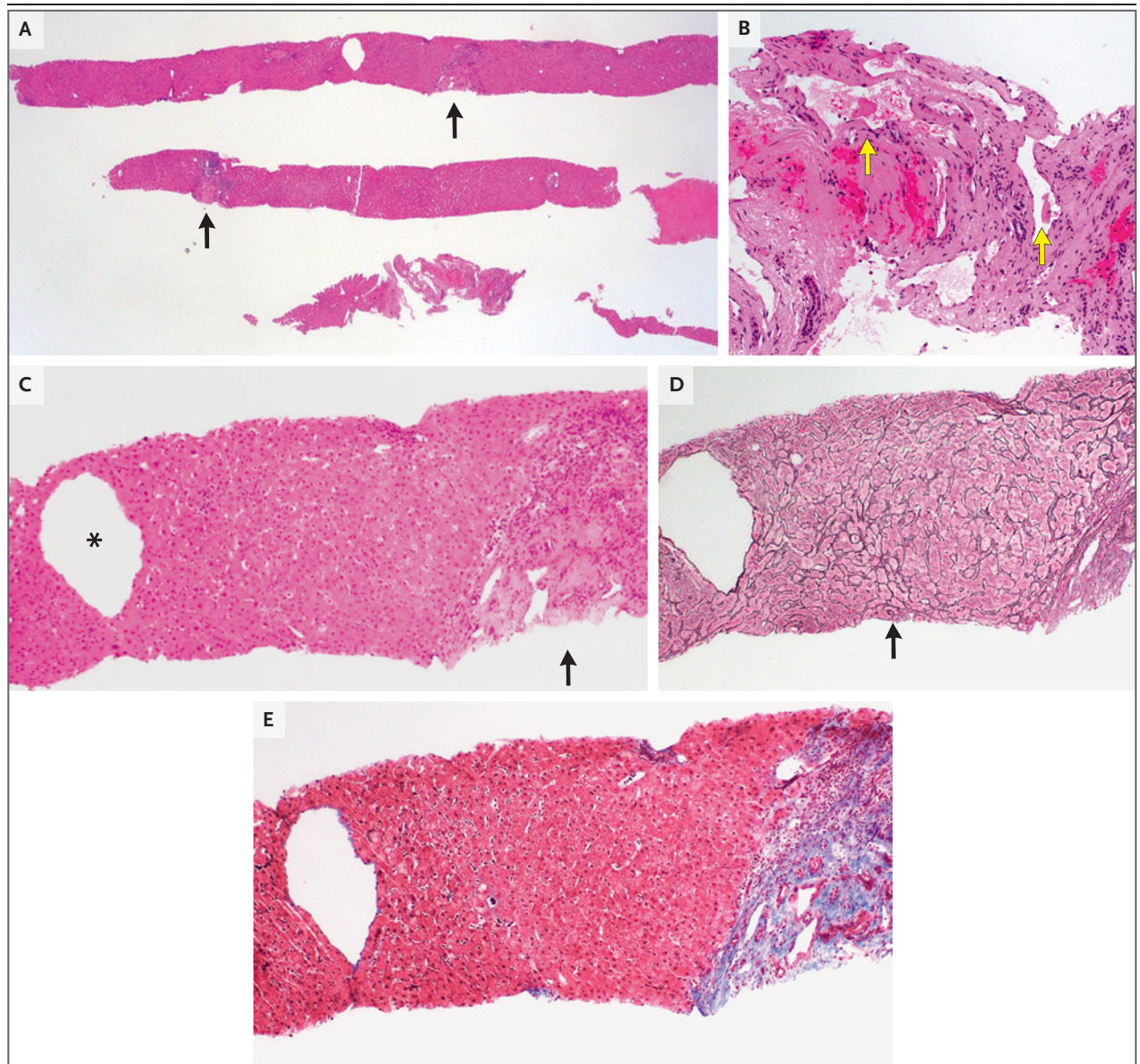


Figure 3. Biopsy Specimen of the Liver.

Hematoxylin and eosin staining of the biopsy specimen obtained from the liver (Panel A) shows portal tracts that are mildly expanded by fibrosis (arrows), but no cirrhosis is present. At higher magnification (Panel B), an expanded portal tract containing multiple portal vein lumens, some with intraluminal fibrin (arrows), is visible. At medium magnification (Panel C), hepatic parenchyma with mild distortion of lobular architecture (the asterisk indicates the central vein) and an expanded portal area with portal vein remodeling (arrow) are visible. Reticulin staining (Panel D) shows a vaguely nodular pattern of expansion with compression and atrophy of hepatic plates (arrow). Trichrome staining (Panel E) shows portal tract expansion (right side) but no evidence of advanced fibrosis or cirrhosis.

splenic veins to improve mesenteric and splenic venous outflow. The risks and benefits were discussed with the patient in detail, and he agreed to the procedure.

The procedure, which was performed under general anesthesia, required obtaining access to the right internal jugular vein to catheterize the middle hepatic vein, obtaining access to the right femoral vein to position the intravascular ultrasound catheter in the suprahepatic inferior vena cava for visualization of the intrahepatic vasculature, and obtaining access to the splenic vein with the trans-splenic approach to recanalize the splenic and portal veins. Venography confirmed chronic venous thrombosis with partial recanalization, diffuse stenosis, and collaterals in the splenic, proximal superior mesenteric, and main portal veins (Fig. 4). Multiple periportal and intrahepatic portal vein collaterals were also noted. The right portal vein was

catheterized, and a small balloon was positioned and targeted for TIPS creation with the use of fluoroscopic and intravascular ultrasound guidance. A TIPS was created between the middle hepatic vein and the right portal vein by placing a stent graft, which was dilated to a diameter of 10 mm. Balloon angioplasty of the main portal vein, superior mesenteric vein, and splenic vein was performed. Venography was performed and confirmed patent superior mesenteric, splenic, and main portal veins with the disappearance of collaterals. A portal venous stent was not placed so that liver transplantation could still be considered if needed.

Dr. Katayoon Goodarzi: Although this patient did not have a personal or family history of venous thrombosis, he was screened for prothrombotic risk factors and received anticoagulation with intravenous heparin followed by treatment with enoxaparin.¹⁹ The hematology service was

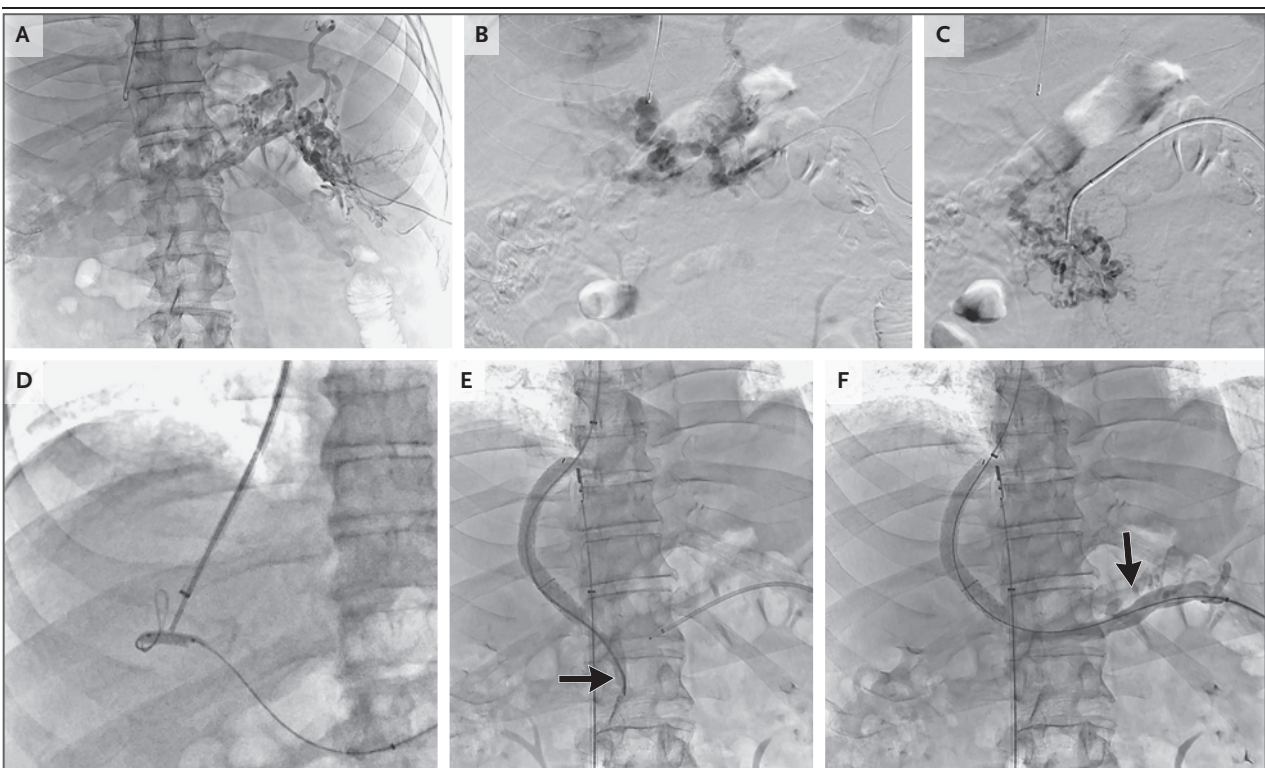


Figure 4. Interventional Radiology Imaging Studies.

Venography of the splenic vein (Panel A), main portal vein (Panel B), and superior mesenteric vein (Panel C) shows diffuse narrowing and collaterals. A transjugular intrahepatic portosystemic shunt (TIPS) was created between the middle hepatic vein and the right portal vein by targeting a balloon placed in the right portal vein (Panel D). Venography performed after TIPS placement and angioplasty shows a patent TIPS (Panels E and F), a patent splenic vein (Panel F, arrow), and a patent proximal superior mesenteric vein (Panel E, arrow), with the complete disappearance of collaterals.

consulted to interpret results of tests for thrombophilia and provide guidance on anticoagulation management.

This patient had normal prothrombin and partial-thromboplastin times. The white-cell count and hemoglobin levels were also normal. These findings would not be consistent with a myeloproliferative neoplasm, the most common systemic cause of noncirrhotic PVT. Results of tests for lupus anticoagulant, anticardiolipin antibodies, and anti- β_2 -glycoprotein 1 antibodies were normal, which ruled out acquired thrombophilia and antiphospholipid syndrome. The patient also underwent hereditary thrombophilia testing, which was negative for factor V Leiden and prothrombin G20210A mutations; levels of protein C and protein S were normal. The antithrombin III activity level was 54% (reference range, 80 to 130) with a proportionally decreased antithrombin III antigen level of 57% (reference range, 80 to 130). A week later, the antithrombin III activity level was 64%, a finding that suggests antithrombin deficiency. However, it was noted that the initial antithrombin III activity level was obtained while the patient was receiving heparin and the repeat antithrombin III activity level while the patient was receiving enoxaparin. Treatment with either heparin or enoxaparin can cause slight decreases in antithrombin III levels through increased clearance. Therefore, assessing an underlying hereditary deficiency is challenging in this patient. I recommended outpatient follow-up to repeat antithrombin III testing, including testing for the V617F mutation in the gene encoding Janus kinase 2 (JAK2) if hereditary antithrombin III is ruled out.

Treatment with warfarin, with an INR goal of 2 to 3, was recommended in this patient with postprandial abdominal pain, nausea, and vomiting due to PVT extension to the superior mesenteric vein resulting in intestinal congestion. Additional considerations included the extension of the clot to the splenic vein and the ongoing possibility of underlying antithrombin III deficiency with a risk of recurrent thrombosis.²⁰ It is important to note that the esophageal varices in this patient were managed with banding and the administration of beta-blockers, and the platelet count remained greater than 50,000 per microliter.²⁰

Dr. Sherman: After the shunt procedure, treatment with lactulose was started to prevent he-

patic encephalopathy. The patient reported that his abdominal pain decreased and he was able to eat a regular diet without any postprandial nausea or vomiting. He was discharged home.

Three months later, the patient reported that he had gained 5.4 kg. He continued treatment with anticoagulation and had no confusion despite stopping treatment with lactulose 3 weeks after discharge. However, the patient noted some mild abdominal pain, and a TIPS interrogation was requested. Unfortunately, the patient missed his appointment for this evaluation, and 6 months after discharge, he reported postprandial nausea with abdominal pain without weight loss. A TIPS evaluation was rescheduled.

Dr. Kalva: Ultrasonography showed slow flow in the TIPS, and TIPS venography confirmed a patent TIPS but recurrent stenosis of the main portal vein, which was treated with angioplasty. The patient underwent follow-up TIPS venography 9 months after discharge, which showed patency of the porto-spleno-mesenteric venous system and the TIPS. CT of the abdomen performed 10 months after discharge confirmed continued patency of these veins and the TIPS.

Dr. Sherman: Despite these interventions, the patient reported continued postprandial nausea with vomiting that occurred 20 to 30 minutes after eating and crampy abdominal pain that lasted several hours. It is well established that portal blood flow increases 30% or more in as little as 15 minutes after a meal, and it is suspected that his symptoms are associated with mesenteric venous congestion after meals.²¹ Ten months after the initial presentation, the patient continues to receive anticoagulation, and there is no evidence of further clot extension. Despite symptoms, his weight is being maintained.

FINAL DIAGNOSIS

Hepatic atrophy due to portal vein thrombosis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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